The Law of Narrative Momentum — Theory

First-Class History, Cost-Conditioned Persistence, and Revision

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Draft — draft-2026-08-30.4
30 August 2026 • document revision 20

DRAFT — draft-2026-08-30.4 r20 (30 August 2026). This document is under active revision and is circulated for comment, not as a finished result. Definitions, numbering and claims may change between drafts; cite the version string above if referring to it. Sections marked as falsifiability commitments are the intended points of attack.

Abstract

This paper develops Narrative as a real, measurable, frame-relative quantity rather than as a metaphorical story stored inside the graph it describes. Graphic Equalisation establishes that a containing frame may possess first-class relational state about a lower graph even where that quantity is absent from the lower graph itself. Realised history can therefore exist as ordinary state in the representing frame.

Narrative is defined as a discrete frame-level measure of realised-history structure that remains operative in the present representation of a graph. The Narrative dimension need not exist in every frame; where it is absent, Narrative is undefined rather than numerically zero. Where history-bearing structure persists across collapse, it enters the next antecedent configuration and thereby biases subsequent reachable geometry toward continuations compatible with that realised structure. Momentum is consequently not introduced by an unexplained grounding arrow from Narrative to a second ontological object. It is the directional dynamical consequence of operative Narrative in a geometry that supplies direction.

The law is stated primarily in *cost* form, which is its primary commitment and requires no directional dimension: surviving history-bearing structure does not remove continuations from the admissible set but prices them, so a continuation that reuses that structure costs nothing further while one requiring it altered must pay the alteration cost. Directional persistence is recovered as the special case for frames that instantiate a direction.

The resulting law is structurally adjacent to established work on hysteresis, path dependence, belief revision, predictive coding, motor schemas, and institutional lock-in, but is not identical to any one of them. The contribution here is the common frame-relative formulation: realised history becomes first-class state, present state carries only the history that remains causally operative, and persistence follows from its presence in the next possibility relation.

1 Dependencies and scope

Reading for the claim. As in the foundational paper, the components used here are individually established and cited as such; the claim is their arrangement, which is not reachable by evaluating those components separately, since they are stipulated to be known. The composition's own falsifiability commitment is stated in the foundational paper and is inherited here.

Graphic Equalisation supplies deterministic realised transition

$$G_t \xrightarrow{\mathcal{C}_t} G_{t+1},$$

and the Seldon-manifold theory supplies counterfactual reachable geometry

$$\mathcal{S}_{h,\rho}(\mathcal{F}_t).$$

This paper adds a measurable higher-frame quantity carrying realised history. It does not require cognition, self-models, language, or deliberative agency.

Hysteresis, path dependence, formal belief revision, and predictive processing are the nearest established structures and are prior art for particular history-dependent behaviours, each treated in Section 14; the present claim is the cross-domain identity of the frame-relative structure underlying them.

2 Narrative as a first-class dimension

Let W individuate graph G.

Definition 2.1 (Narrative dimension)**.** *The Narrative dimension $D_{\mathcal{N},W:G}$ exists in frame W when W represents a history-bearing relation concerning G whose value is causally operative in subsequent transition or reachability structure.*

Before that relation exists,

$$D_{\mathcal{N},W:G} = \text{NULL},$$

and $\mathcal{N}_{W:G,t}$ is undefined. This is the dimension/value typing of Graphic Equalisation, stated in full in Seldon Manifolds: NULL is not the value zero, and zero is a legitimate value once the dimension exists.

Definition 2.2 (Narrative)**.** *Where $D_{\mathcal{N},W:G}$ exists,*

$$\boxed{\mathcal{N}_{W:G,t} \in \mathbb{Z}}$$

is the discrete frame-relative measure of realised-history structure concerning G that remains represented and operative in W at realised step t.

The integer is a state index rather than a globally fixed unit. A grounding may map it to arbitrary dimensional scales or dynamically varying quantisation.

Proposition 2.3 (No emergence gap)**.** *The absence of Narrative as a primitive in $G = (g, r, \Gamma)$ does not make Narrative non-physical or metaphorical. Where the defining relation exists in W, Narrative is ordinary first-class state in W.*

This is a direct use of the higher-frame principle established in Graphic Equalisation: a relation may be absent from the lower graph while being fully real in a frame containing the relation required to define it.

3 Narrative and realised history

Realised history is the represented succession

$$\mathcal{H}_{W:G,\leq t} = G_0 \xrightarrow{\mathcal{C}_0} G_1 \xrightarrow{\mathcal{C}_1} \cdots \xrightarrow{\mathcal{C}_{t-1}} G_t.$$

Narrative is not necessarily the complete history. It is the currently operative history-bearing structure:

$$\mathcal{H}_{W:G,\leq t} \longrightarrow \mathcal{N}_{W:G,t}.$$

Proposition 3.1 (Historical sufficiency of present structure)**.** *Past events affect future dynamics only insofar as their consequences remain represented in present graph or boundary structure.*

This proposition is closely related to the state-variable treatment of systems with memory: the past influences future behaviour through whichever memory variables remain operative in the present representation. Hysteresis theory makes this explicit in systems whose present response retains selective information about past extrema or branches (Mayergoyz, 1991, 2003). The present theory generalises the statement across frames and graph types.

Narrative is therefore not "everything that happened." It is the part of realised history that has not disappeared dynamically.

4 Direction and momentum

Narrative magnitude alone is not a vector. Direction exists only where the relevant frame also instantiates a directional dimension or ordering over continuation.

Let $\hat{\mathbf{d}}_{W:G,t}$ denote a locally represented direction associated with the history-bearing relation. In a grounding where scalar Narrative magnitude and directional coordinates are commensurable, a convenient coordinate representation is

$$\boxed{\mathbf{p}_{W:G,t} = \phi_W(\mathcal{N}_{W:G,t})\,\hat{\mathbf{d}}_{W:G,t},}$$

where ϕ_W supplies the frame-specific scale. The special case $\phi_W(n) = n$ gives $\mathbf{p} = \mathcal{N}\hat{\mathbf{d}}$.

Definition 4.1 (Narrative momentum)**.** *Narrative momentum is the directional persistence in reachable future geometry induced by operative Narrative when a directional dimension exists in the frame.*

Momentum is therefore derived from first-class Narrative plus represented geometry; it is not posited as a second unexplained substance. Classical momentum is one exact grounding of this structural form.

5 Momentum is present structure

The word "momentum" invites a reading this theory does not take, and the law below turns on rejecting it.

Proposition 5.1 (Momentum is not history carried through time)**.** *Momentum is a* currently instantiated structural consequence *of prior change, which constrains what can happen next:*

$$\boxed{\textit{prior change} \;\rightarrow\; \textit{present momentum} \;\rightarrow\; \textit{determinable continuation.}}$$

The past need not continue to exist for momentum to be real. What is real is the structure prior change left behind, instantiated now.

This is Graphic Equalisation's history principle applied to a specific quantity: history matters only where it survives as present structure. A momentum term is exactly such surviving structure.

Remark 5.2 (Momentum is the un-rewritten Narrative; to change course is to change it)*.* What a frame carries forward is not a bare delta but its operative Narrative — the surviving history-bearing structure S of Hypothesis 6.1. The Narrative is not pointed at through any indirection; it is the frame's own standing structure, and a delta is a write *to* it — a self-change of S, gated by the container (Graphic Equalisation) — not a separate thing the frame carries. Continuation is then not a repeated action but the absence of one: nothing re-applies S each step, because there is no step — no global tick — for it to be re-applied at (Graphic Equalisation, collapse as a local write). S persists until a rule writes against it, and that write is the only priced

operation. This is why the "Nothing pushes" reading (Remark 6.3) is exact rather than a manner of speaking: persistence is the free default and change the paid exception, because retaining the Narrative destroys nothing and altering it does (Graphic Equalisation, Remark on evaluation as a write). Put as a definition: the Narrative is what a subgraph runs when *no rule fires* — the free continuation in its pure form, S simply carrying forward with nothing writing against it. Having rules adds only the *possibility* of a write; the Narrative is what is left running when none occurs. And the Narrative is relative in the reading, not in the carrying: the *internal* Narrative — the frame's own — is S read in the terms its current rule set fixes, what that rule set could write against, while an *external* Narrative is another frame's cross-frame reading of the same structure (the "velocity" below). Change the current rule set and the internal reading changes with it, since the terms have changed; what is read is still S, and there is no rule-set-independent *reading* of it. To change course is therefore, literally, to change the Narrative — a write to S — and its price is the alteration cost C_{alt} of Hypothesis 6.1, which in the mechanical grounding is the impulse $J = \Delta\mathbf{p}$ (Remark 10.1).

It also fixes what the momentum quantity is. Read from the Narrative, it is a difference, Δ, and not a difference per unit time, $\Delta/\Delta T$: the divisor is the inferred time dimension, which is not available in-frame (Graphic Equalisation, the missing denominator). "Velocity" is therefore an observer's cross-frame reading of the Narrative's persisting difference, not the substrate quantity, and the substrate statement of the first law is that the Narrative continues un-rewritten until a write — a change of the Narrative — changes it.

Remark 5.3 (The Narrative level stack: inferred, carried, read)*.* Narrative is frame-relative for a reason it inherits rather than assumes: Graphic Equalisation's founding axiom places every distinction and represented quantity in some frame, and a history-bearing measure is such a quantity, so a frame-free Narrative would be the view from nowhere that axiom denies. Because Narrative is frame-relative, then, it is stacked, and the levels differ not only in whose trajectory they describe but in *how a frame can come to know them at all.* Three levels are distinguishable from any given frame.

Narrative $+1$ — *inferred.* The container's own trajectory is not readable from inside. What reaches the contained frame is only what the container *does to it*: deformations arriving from above, which it did not author. A frame therefore reconstructs its container's Narrative abductively, from the imposed changes it registers, and never by inspection. This is the same asymmetry Graphic Equalisation gives between container and contained — the operation is the self-similar one in both directions, but representational *capacity* is not symmetric, and a subframe cannot hold its container whole. Inferring $+1$ is what observation is for.

Narrative 0 — *carried.* A frame's own Narrative is neither read nor inferred but simply carried: it is the standing structure S itself, not pointed at through any indirection, and the frame's trajectory through its own reachable geometry. Being inside it is what makes it inaccessible as an object; seeing one's own trajectory *as* a trajectory requires either an external frame or a self-model built for the purpose.

Narrative -1 — *read.* Subgraphs a frame has reflective access to are readable where it instantiates the relations to read them; their direction is available to it, where it also instantiates a directional dimension, in the way its own is not.

None of the three is compulsory. The Narrative dimension may be NULL in any frame (Definition 2.1), and each level requires its own machinery: inference to reach $+1$, a self-model to objectify 0, reflective access to read -1. A frame may have any, all, or none of these. What is automatic is only the *carrying*, and that only because persistence is free; representing what is carried is a further and optional matter.

One consequence is worth stating, since it is often mistaken for a difference in difficulty. Inferring the container and reading a subgraph are different *modes*, not harder and easier instances of one activity — which is why an account built for one does not transfer to the other.

6 The Law of Narrative Momentum

The law is stated first without reference to direction, because Graphic Equalisation defines none. Realised collapse induces an *order* over events; it does not supply a basis, an orientation, or a vector space. Direction is a frame's optional instrumentation (Section 4), and a law that presupposed it would be a claim about representations rather than about reachability. The directional statement is recovered below as the special case for frames that instantiate a directional dimension.

Hypothesis 6.1 (Law of Narrative Momentum, cost form)**.** *Let S be operative history-bearing structure that survives a realised transition and is materially unchanged at $t+1$. Then for each admissible continuation $a \in \mathcal{P}(G_{t+1})$, the cost of a is conditioned on S:*

$$C(a \mid S) = \begin{cases} C_0(a) & a \text{ reuses } S, \\ C_0(a) + C_{\text{alt}}(S, a) & a \text{ requires } S \text{ altered or replaced,} \end{cases}$$

where C_{alt} is the cost of the interaction that alters S.

Remark 6.2 (What is conditioned, and what is not)*.* Γ **decides membership; Narrative decides cost.** Nothing leaves the admissible set because of history. The set is what the admissibility grammar admits, and surviving structure prices the members. This division is what keeps the two quantities separable: an account that let history remove continuations would have Narrative doing Γ's work, and no way to tell which was responsible for an absence.

Remark 6.3 (Nothing pushes)*.* The hypothesis posits no force, no tendency, and no stored impetus. The whole asymmetry is that *continuing costs nothing and altering costs*: S is already instantiated, so a continuation employing it requires no further expenditure, while one contradicting it must pay C_{alt}. Persistence is what an unaltered structure looks like from the next step, not something acting on the graph.

Remark 6.4 (Irreversibility is a limit case, not a removal)*.* Where $C_{\text{alt}}(S, a)$ exceeds any expenditure available in the frame, a is unreachable in practice while remaining admissible. From inside the frame this is indistinguishable from inadmissibility; it is different in kind, and conflating them relocates a budget fact into the grammar. Irreversibility is therefore frame- and budget-relative rather than absolute, consistent with the companion manifold paper's treatment of it as a property of a deformation rather than of an act.

Corollary 6.5 (Material redirection)**.** *Redirection therefore requires paying C_{alt} — an interaction that changes the relations carrying S. It is not achieved by preference over continuations, by representing a different continuation, or by intending one: those leave S instantiated, so the price stands and the cheap continuations remain the ones that reuse it.*

Corollary 6.6 (Directional form)**.** *Where frame W instantiates a directional dimension and $\mathcal{N}_{W:G,t}$ is operative, the cost conditioning of Hypothesis 6.1 projects into that dimension, and subsequent reachable geometry retains a directional component aligned with the continuation the Narrative represents — because the cheapest continuations are those reusing S. This is the statement in directional language, and it holds only where the frame supplies the direction. Two frames may instantiate opposed directional dimensions over the same S and disagree about the sign of the retained component while agreeing on the conditioning that produces it.*

Remark 6.7 (Why the cost form is the substrate-level statement)*.* Under the directional statement alone, a frame with no directional dimension makes the law vacuous, which would be a strange property for a claim about what is reachable. The cost form applies wherever a possibility set exists, requires no basis, and is invariant under the choice of one — a gauge condition of the

same kind the companion construction paper imposes on relative weighting. A worn route does not possess a direction, it possesses differential cost, and two travellers moving oppositely along it experience opposite directions from identical structure.

The key recursion is

history survives as structure ⇒ history enters the next possibility relation.

The law itself is deterministic. For empirical work over repeated cases where the observer does not represent every determining dimension *and instantiates a directional dimension*, so that Corollary 6.6 applies, an ensemble statistic such as

$$\frac{1}{m}\sum_{i=1}^{m}\mathbf{p}_{t+1}^{(i)}\cdot\widehat{\mathbf{p}}_{t}^{(i)}>0$$

may test directional persistence across cases. The statistic measures the observer's population-level evidence; it does not add stochasticity to the Graph Law.

7 Narrative revision

Definition 7.1 (Narrative revision)**.** *A material Narrative revision is a realised change*

$$\Delta\mathcal{N}\neq 0$$

large enough to alter future distinctions or direction at the operative frame and resolution.

A proposition may change while Narrative does not. Conversely, a small local change may induce a large Narrative revision when it alters a highly connected history-bearing relation.

Formal belief-revision theory already distinguishes local proposition change from revision of a larger belief set and studies the conditions under which contraction or revision preserves as much prior structure as possible (Alchourrón et al., 1985). The present framework differs by treating the relevant dependency structure as dynamical graph state rather than purely propositional closure, but the structural predecessor is direct.

8 Structural lock-in

Hypothesis 8.1 (Structural lock-in)**.** *For fixed frame and revision class, the minimum intervention required for material Narrative revision tends to increase with the amount of present relational structure that must be modified, bypassed, or rendered irrelevant.*

If $K_R(\mathcal{N})$ is a discrete measure of Narrative-supporting structure, an appropriate monotone grounding predicts that increasing K_R does not decrease minimum revision cost after relevant capacity and geometry are controlled.

This generalises familiar forms of lock-in. In technology and economics, increasing returns can make an historically selected path progressively harder to leave (Arthur, 1989; David, 1985). In institutional analysis, timing, sequence, and increasing returns can similarly make established arrangements difficult to reverse (Pierson, 2000). These are groundings of the structural-lock-in hypothesis, not its proof.

9 History-sensitive projection

Let H_t be dynamically relevant history and $\widehat{H}_t$ represented history.

Hypothesis 9.1 (History-sensitive projection)**.** *Where two represented histories imply different operative present relations despite selected instantaneous observables matching, their projected reachable geometries diverge over some relevant horizon:*

$$\widehat{H}_t \neq H_t \Rightarrow \widehat{\mathcal{S}}_{h,\rho}(\mathcal{F}_t) \neq \mathcal{S}_{h,\rho}(\mathcal{F}_t).$$

The load-bearing clause is "different operative present relations." If the complete causally operative present state is identical, Graphic Equalisation is deterministic and the histories cannot produce different successors merely because they were different histories. History matters only where it survives in state, exactly as Proposition 3.1 requires.

Predictive-coding models provide a cognitive analogue: higher-level predictive state shapes the processing of subsequent input, while residual error updates that state (Rao and Ballard, 1999). Motor-control schema theory likewise treats prior movement experience as modifying generalized structures used to parameterise later action (Schmidt, 1975). Neither is claimed to be identical to Narrative Momentum; both demonstrate established domain-specific cases in which retained history changes future projection.

10 Mechanical grounding

Classical momentum is a constrained quantitative grounding. In a mechanical frame, realised positional history together with a time embedding determines velocity; mass supplies the established mechanical scale and

$$\mathbf{p} = m\mathbf{v}.$$

The present theory identifies the same abstract ingredients—frame-relative history-bearing state, direction, persistence, and redirection—in a domain where the quantitative law is already exact.

Remark 10.1 (Under the cost form the correspondence is an identity)*.* Newton's first law is the statement that continuing costs nothing. Altering motion requires impulse, and the impulse needed to bring a body to rest is

$$J = \Delta\mathbf{p} = m\mathbf{v},$$

so *the momentum of a body is exactly the cost of stopping it.* Hypothesis 6.1 is that sentence with the currency left unspecified: C_{alt} in the mechanical grounding is impulse, and mass is the exchange rate converting a desired change of motion into required expenditure.

Remark 10.2 (Two currencies for one alteration)*.* The same alteration has two established mechanical prices: $m\mathbf{v}$ in impulse and $\frac{1}{2}mv^2$ in energy. They are not proportional, and neither is more correct. This is a concrete instance of the commitment that costs need not share a unit or an origin across frames or classes — a point that otherwise reads as a hedge, and which classical mechanics already demonstrates within a single grounding.

Remark 10.3 (A grounding, not an analogy)*.* The correspondence should not be described as analogical. Ordinary physical momentum and Narrative momentum have the same structural form under Proposition 5.1 — prior deformation leaves present structure that constrains reachable continuation — and differ only in which relations carry that surviving structure, represented rather than material. Under Axiom −1 an inertial reference frame and a representational frame are the same kind of object — a locus at which distinctions are made, with quantities indexed to it and no view from nowhere available. They differ in how much structure is instantiated, not in kind. A mechanical grounding is therefore an *instance* of the present claims in a frame carrying a great deal of structure, and calling it an analogy would reintroduce the privileged-physical / derivative-representational division that Axiom −1 removes.

What must not be overread is *scope*, which is a different and real constraint: Narrative is not asserted to carry SI momentum units, non-mechanical groundings need not obey conservation of mechanical momentum, and no claim is made that a non-mechanical grounding inherits any mechanical law beyond the structural form stated in Hypothesis 6.1.

11 Biological grounding: evolutionary momentum

The mechanical grounding of Section 10 is the clearest but not the most useful, since impulse is where the intuition came from. A grounding is worth more where the construct was named independently, by people with no contact with this account, and where it has been *measured*. Population genetics supplies one, under the same word.

Where a character is transmitted by maternal inheritance, a population continues to change after selection for many generations after selection has ceased, and the response can run in the direction selection opposed; rate and direction change without any change in the strength of selection or the parameters of inheritance (Kirkpatrick and Lande, 1989). The effect is named *evolutionary momentum* in the follow-on treatment (Lande and Kirkpatrick, 1990), and it has been measured in a wild population, where responses in red squirrels are influenced by selection in the previous generation (McAdam and Boutin, 2004). Rate itself is handled as an inherited, autocorrelated quantity elsewhere in the same field (Thorne et al., 1998).

What that grounds is Section 5 entire. A lineage's continued change after the forcing stops is not a memory of past selection acting at a distance — there is no mechanism for that — but present structure carrying forward, which is exactly the claim that momentum is present structure rather than stored history. The maternal contribution is the standing structure, the continued response is the free continuation, and a generation in which the direction reverses is the case where following the standing structure costs less than writing against it.

Remark 11.1 (The lag is a description, not a rival mechanism). The published account describes the effect as a *lag*: the response persists because the maternal contribution is one generation behind. An earlier version of this section treated that as a competing mechanism and staked a falsifier on distinguishing the two. It is not a competitor, and the reason is the foundational paper's.

Transmission cannot lag, because there is no time for it to lag in. A lag is a delay, and a delay requires a time dimension for the delay to occur in — which the substrate does not supply (Graphic Equalisation, on time as inferred rather than primitive). What is available beneath it is causal order, and in causal order "lags by a generation" says only that the effect appears at the next position. That is a *location* and not a mechanism: it states where the continuation is observed and not why there is one.

Structurally there is no delay at all. An offspring's present structure includes a contribution assembled from its parent's, and it acts on what it holds, now. Nothing is catching up. The appearance of lateness belongs to an observer indexing generations against an external clock, and the temporal metaphor — *the system is still responding to selection that has stopped* — borrows explanatory force it does not have, since nothing is responding to anything absent; a structure is carrying forward.

The published account describes the effect as a *lag*, and two earlier versions of this section treated that as something to correct — first as a rival mechanism, then as a description that a finer aperture would dissolve. Both were wrong, and the reason is worth stating rather than quietly dropping, because it is the more interesting fact.

Kirkpatrick and Lande hold the position this paper holds. Their carrier is the maternal phenotype, written into the model as a matrix acting on the mother's trait values — present structure, in the offspring's own generation, doing the work. Nobody in that literature claims past selection acts at a distance, and there is no mechanism there for it to. So the argument this section previously mounted was aimed at a position no one occupies, and the correction was agreement dressed as a correction.

One claim from those earlier versions survives and is worth keeping at its proper size, because it is one sentence rather than the argument that was built on it. *Nothing can be pending that is not currently held.*

Pending states are real and this does not deny them. What it denies is pending in the void

— a system in a state of waiting with nothing instantiating the wait. Every case offered as a counterexample is an instance: a stalled instruction is held in the pipeline that stalled it, an outstanding cache miss is held in the register tracking it, a signal in flight is held in the medium carrying it, and the maternal contribution is held in the mother. In each the waiting has an address. A delay is not an interval of nothing happening; it is structure, somewhere, for the duration — which is why the causal-set treatment cited above defines an order-interval as the *set of elements* between two others rather than as a gap, and takes its measure by counting them.

That is also the same requirement the foundational paper's economics makes of a read: what is returned has to be somewhere, and there is no channel outside the structure for it to be in. A pending state with no holder would be information existing nowhere.

Falsifier. Fails if a system can be exhibited in a pending state — a determinate outcome not yet realised, whose realisation depends on nothing further arriving — with no structure anywhere holding the pendency. Exhibiting that the holder is not *resolved* by a particular observer is not such a case.

What the case supplies is therefore an illustration and not a repair: a carrier that is written down, parameterised and estimated, in a domain with no contact with this account, doing what Section 5 says a carrier does. That is what a grounding is for, and it is enough.

Two things it does not supply, which the previous versions claimed. It does not test the cost form, which the abstract calls this paper's primary commitment: there is no cost quantity anywhere in the cited work, and describing a sign change in the maternal-effect matrix as "following the standing structure costs less than writing against it" redescribes rather than measures. And the model damps: the response decays by the leading eigenvalue of that matrix each generation, so the standing structure is rescaled rather than materially unchanged, which is not the free continuation of Hypothesis 6.1 but a decaying recursion. Whether that damping is a failure of the antecedent or a case the hypothesis should be widened to cover is open, and is the sharpest question the biological case puts to this paper.

None of which makes the published treatment wrong. Quantitative genetics is a time-indexed model by construction and the lag description is exact within it. What this account observes is that the mechanism does not require the index: the same effect, described without a time dimension, is present structure carried forward, and that is what makes it a grounding for Section 5 rather than an analogy to it.

Two limits. This is not the claim that evolutionary momentum *is* Narrative momentum under another name — it is one mechanism in one domain, and the shared word is a fact about vocabulary rather than an argument. And by the composition commitment of the foundational paper, an independent re-derivation of this kind evidences the constraints that force it rather than the arrangement this account proposes, so none of it is offered as confirmation.

12 Cognitive and institutional groundings

In cognition, Narrative-supporting relations may include expectations, habits, motor programmes, autobiographical commitments, causal explanations, or identity-bearing relations; predictive coding, motor-schema theory, and formal belief revision supply the established cases (Section 14).

In institutions, contracts, procedures, staffing, compatibility commitments, infrastructure, capital allocation, and reputation can carry Narrative, with path-dependence research as the direct precedent (Section 14). Routine output may therefore remain stable even when the institution's capacity to materially revise the structure carrying that routine has degraded.

These are groundings of the abstract structure.

13 Relationship to Narrativium

Narrative Momentum describes persistence and Narrativium capacity for deformation: $\mathcal{N}$ is operative history-bearing structure, N the reachable capacity to change it, and the companion Narrativium paper sets out that contrast and its counterpart against Muchness. Revision horizons produced by comparing current capacity with minimum revision cost are developed separately.

14 Positioning and prior art

The core ingredients of Narrative Momentum are individually old. Hysteresis theory formalises selective memory in dynamical systems (Mayergoyz, 1991, 2003). Path dependence and increasing returns formalise historically conditioned lock-in in economics and institutions (David, 1985; Arthur, 1989; Pierson, 2000). AGM theory formalises dependency-sensitive belief revision (Alchourrón et al., 1985). Predictive coding and motor-schema theories formalise history-bearing internal structure affecting future processing and action (Rao and Ballard, 1999; Schmidt, 1975).

The claimed contribution is the reduction rather than the ingredients: these phenomena can be expressed as instances of one frame-relative pattern,

> realised history → operative present structure → cost-conditioned persistence of reachable continuations.

Graphic Equalisation supplies the causal reason the pattern works: only present structure enters the next deterministic transition relation. Seldon manifolds supply the geometry in which that persistence is expressed. Narrative names the measurable higher-frame state that carries the surviving history.

15 Falsifiability commitments

On the tags. Each commitment below is marked [SHAPE] where failing it costs this grounding its *reach* — the account would no longer apply to the domain, or would apply only by giving up what it was brought here to explain — and [DETAIL] where failing it costs a specific mechanism while the grounding stands. The substrate paper has carried these marks for some time and the groundings did not, which had the discipline applied where the risk is smallest: the substrate risks little by design and the groundings carry the exposure. They were assigned in a single pass under the rule just stated, and are the least-audited judgement in this document — a reader who thinks one is marked in the flattering direction is disagreeing with a fast call, not an argued position.

1. [SHAPE] **First-class Narrative.** Fails for a claimed grounding if Narrative cannot be measured as frame-level state and exists only as an after-the-fact interpretive label.

2. [SHAPE] **Discrete Narrative.** Fails if Narrative-dependent observable behaviour requires a non-discrete primitive that no dynamically quantised discrete representation can preserve.

3. [SHAPE] **Historical sufficiency.** Fails if erased history with no present causal trace systematically changes future transition structure.

4. [SHAPE] **Narrative Momentum (cost form).** Fails for a specified system class if a continuation requiring operative history-bearing structure to be altered carries no greater cost than one reusing it, absent any interaction that alters that structure. This is the primary commitment, and it is testable without a directional dimension. It also fails if history is found to remove continuations from the admissible set rather than to price them, which would place the effect in Γ (Remark 6.2).

5. [SHAPE] **Narrative Momentum (directional form).** Fails for a frame instantiating a directional dimension if operative Narrative-supporting structure and its represented direction persist while the associated directional tendency repeatedly vanishes or reverses without a relevant interaction. Failure here without failure of the cost form indicts the frame's directional instrumentation rather than the law.

6. [DETAIL] **Momentum without a carried rate.** Fails if every measured case of continued change after forcing ceases is fully accounted for by a decaying recursion on inherited state — the published maternal-effect form — with no case in which a rate persists undamped in the way Hypothesis 6.1 requires. The cited biological cases are damped, so on present evidence this commitment is closer to failing than to holding (Remark 11.1).

7. [DETAIL] **Structural lock-in.** Fails if increasing the structure that must be revised systematically lowers minimum revision cost after relevant capacity and geometry are controlled.

8. [SHAPE] **History-sensitive projection.** Fails if histories that produce different operative present relations never produce different reachable projections in the claimed domain; conversely, the hypothesis is misapplied if different histories with identical causally complete present state are claimed to produce different deterministic successors.

9. [DETAIL] **The Narrative level stack.** Fails if a frame can read its container's Narrative directly rather than inferring it from imposed change, or if reading a subgraph proceeds abductively from imposed change in the same way inferring a container must. (No claim is made that any level is compulsory: each may be NULL, and the modes are claimed to differ in kind, not in difficulty.)

16 Conclusion

Narrative is not a metaphor smuggled into dynamics. It is first-class state in the frame that contains the relations from which it is measured:

$$\boxed{\text{collapse} \to \text{realised history} \to \mathcal{N} \to \text{cost-conditioned persistence of reachable continuations.}}$$

Momentum no longer requires a magical bridge. Narrative is already dynamical state in the higher frame; the law is the cost conditioning that state imposes on subsequent reachable geometry, and the directional persistence the name suggests is the special case for frames that instantiate a direction (Corollary 6.6).

The constituent ideas have long precedents. The proposed novelty is that the same minimal recursive structure accounts for their common form without making Narrative a primitive of every graph or treating history as a force that acts from the past without surviving present structure.

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